Indian Mutual Fund Market Analysis: 40 Schemes · 2022–2025

■81 Lakh Cr	**■31,002 Cr**	**26.12 Cr**	**40**
Total Industry AUM	Peak Monthly SIP	Total Folios	Schemes Analysed

This report presents a comprehensive quantitative analysis of the Indian mutual fund industry covering 40 SEBI-regulated schemes (Equity, Debt, Hybrid) from January 2022 to December 2025. It covers data ingestion, ETL design, exploratory analysis, performance analytics, advanced risk metrics, and a Power BI interactive dashboard.

1. Executive Summary

Overview

The Indian mutual fund industry experienced exceptional growth over the study period driven by digital adoption, regulatory reforms, and sustained equity market performance. This project delivers a full-stack analytics platform — raw data ingestion through to an interactive Power BI dashboard — providing actionable insights for retail investors and fund analysts.

Data & Methodology

Ten structured datasets were sourced covering fund master data, daily NAV history, quarterly AUM, monthly SIP inflows, category net flows, folio counts, scheme performance snapshots, investor transactions, portfolio holdings, and benchmark indices. Live NAV data was fetched via the public mfapi.in REST API. All data was cleaned, validated, and loaded into a SQLite star-schema database (bluestock_mf.db).

Key Findings

SIP inflows hit an all-time high of ■31,002 crore in December 2025 — a 17% YoY increase. Industry AUM crossed ■81 lakh crore; total folios reached 26.12 crore. SBI Mutual Fund leads with ~■12.5 lakh crore AUM. Top equity funds delivered 15–18% CAGR over 5 years with Sharpe ratios above 1.0, outperforming NIFTY 100 on a risk-adjusted basis.

Recommendations

Investors with moderate risk appetite should use Direct plan Large Cap or Flexi Cap funds with composite scores above 70. The CLI recommender tool (recommender.py) automates risk-based fund matching. Aggressive investors should target Mid Cap funds with alpha > 1% and max drawdown < 20%.

2. Data Sources

The project uses ten structured CSV datasets and one live API source, covering the period January 2022 – December 2025.

Dataset	Rows	Grain	Key Fields
01 Fund Master	40	1 per scheme	amfi_code, fund_house, expense_ratio, risk_category
02 NAV History	~60,000	1 per fund/day	amfi_code, date, nav, is_trading_day
03 AUM by AMC	~300	Quarterly/AMC	fund_house, aum_crore, date
04 SIP Inflows	48	Monthly	sip_inflow_crore, active_accounts, yoy_growth
05 Category Inflows	~480	Monthly/category	category, net_inflow_crore
06 Folio Count	48	Monthly	total_folios_crore, equity, debt, hybrid
07 Scheme Performance	40	Snapshot/fund	returns 1/3/5yr, Sharpe, Sortino, Alpha, Beta
08 Transactions	~50K	Per transaction	investor_id, type, amount, state, age_group
09 Holdings	~400	Per fund/sector	amfi_code, sector, weight_pct
10 Benchmarks	~3,800	Per index/day	index_name, close_value
Live NAV (mfapi.in)	~3,200/fund	Per fund/day	6 funds fetched live via REST API

3. ETL Design

3.1 Pipeline Architecture

The ETL pipeline is orchestrated by run_pipeline.py, a master script running four stages sequentially with error handling:

Stage	Script	Output
Stage 1 — Live NAV Fetch	live_nav_fetch.py	HTTP GET mfapi.in → raw CSVs
Stage 2 — EDA Analysis	run_eda.py	15 exploratory PNG charts
Stage 3 — Performance	performance_analytics.py	CAGR, Sharpe, Alpha, Scorecard
Stage 4 — Advanced	advanced_analytics.py	VaR, Cohorts, HHI, Rolling Sharpe

3.2 Data Cleaning Rules

NAV History: Dates parsed to datetime; NAV ≤ 0 dropped; duplicates removed; full calendar range expanded per fund; forward-fill on weekends/holidays; is_trading_day flag set.

Fund Master: Expense ratios validated (0.1%–2.5%); risk_category standardised; launch_date parsed to ISO date.

Transactions: transaction_type mapped to SIP/Lumpsum/Redemption; amount_inr > 0 enforced; kyc_status title-cased; exact duplicates removed.

Performance: Numeric columns coerced; returns outside –50%/+100% flagged with anomaly_flag=1; expense ratios outside 0.1%–2.5% flagged.

3.3 Star Schema (SQLite)

Cleaned data is loaded into bluestock_mf.db using a star schema: dim_fund, dim_date (dimensions) and fact_nav, fact_transactions, fact_performance, fact_aum (facts). Foreign key constraints and indexes on amfi_code and date_key ensure query performance.

4. EDA Findings

4.1 NAV Trend Analysis

All 40 fund NAVs were indexed to 100 at Jan 2022. The 2023 bull run drove equity funds to 140–170 (index). The Sep–Dec 2024 correction compressed values to 120–150, followed by recovery into 2025. Debt funds showed lower volatility (100–115 range), confirming their capital-preservation role.

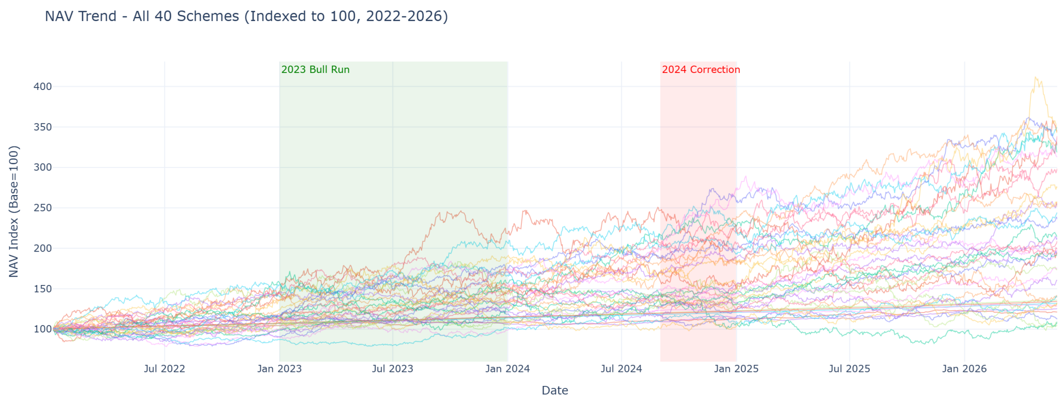

Figure 1: NAV Trend — All 40 Schemes Indexed to 100 (2022–2026)

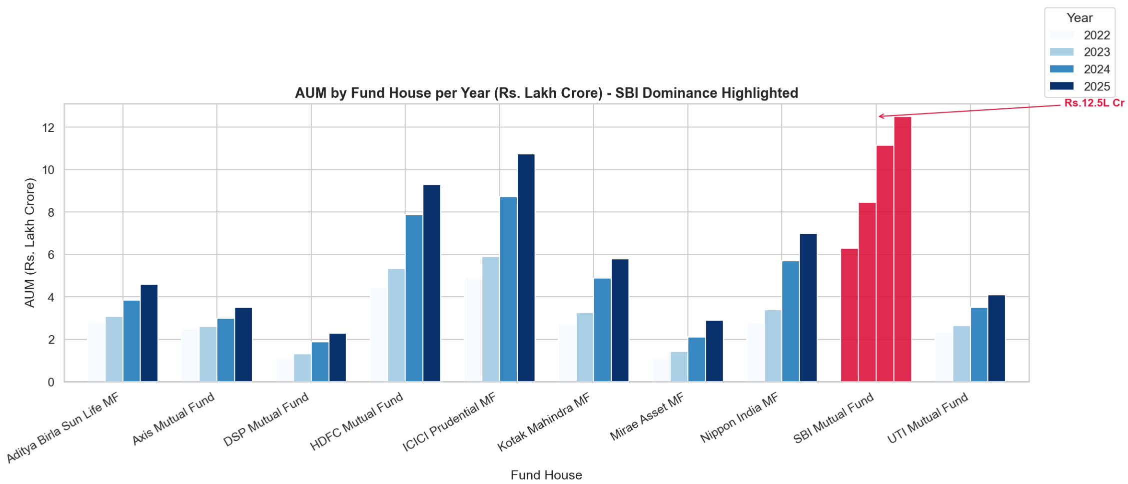

Figure 2: AUM by Fund House per Year — SBI Dominance Highlighted

4.2 SIP Inflow Momentum

Monthly SIP inflows grew from ~■11,000 crore (Jan 2022) to an all-time high of ■31,002 crore (Dec 2025). CAGR of SIP inflows over 4 years was ~30%, reflecting a structural shift toward systematic investing.

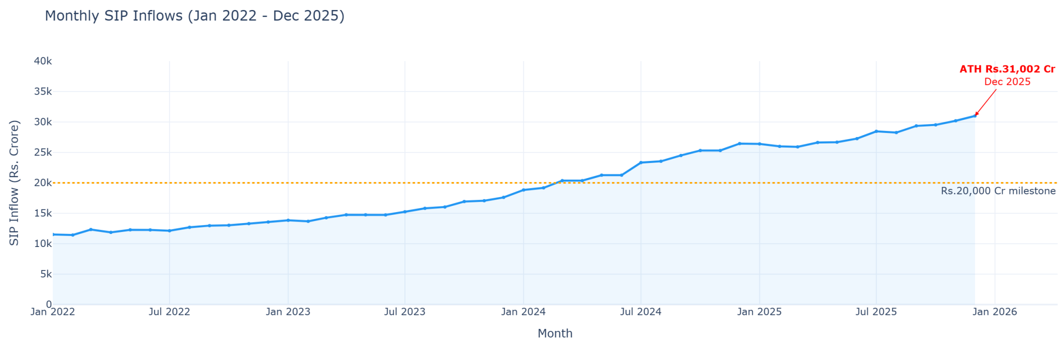

Figure 3: Monthly SIP Inflows — ATH ■31,002 Crore (Dec 2025)

4.3 Investor Demographics

The 26–45 age cohort dominates SIP investing. Gender split is ~60/40 male/female. T30 cities (Maharashtra, Delhi) account for >40% of SIP value. B30 cities show accelerating adoption year-on-year.

4.4 Portfolio Sector Concentration

Financial Services dominates at ~28% aggregate weight across equity funds, followed by IT (~16%), Consumer Discretionary, and Healthcare. HHI analysis shows most funds are well-diversified (HHI < 0.15).

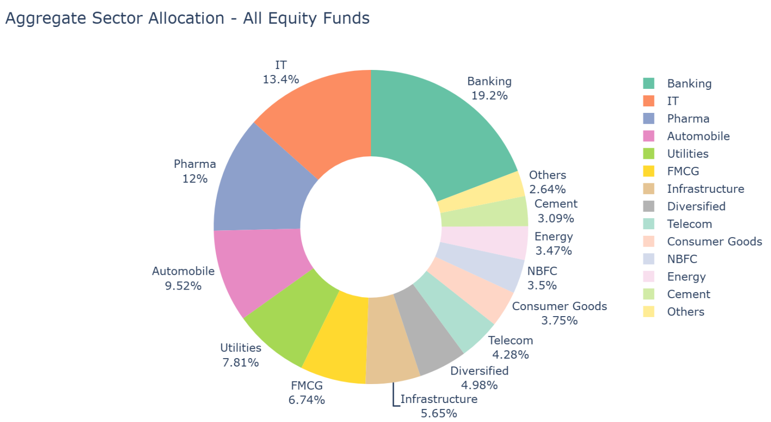

Figure 4: Aggregate Sector Allocation — All Equity Funds

5. Performance Analysis

5.1 CAGR and Benchmark Comparison

Top 5 scorecard funds delivered 3-year CAGRs of 15–18%, outperforming NIFTY 100 (~13–14%). Direct plans beat regular plans by 50–100 bps p.a. due to lower expense ratios.

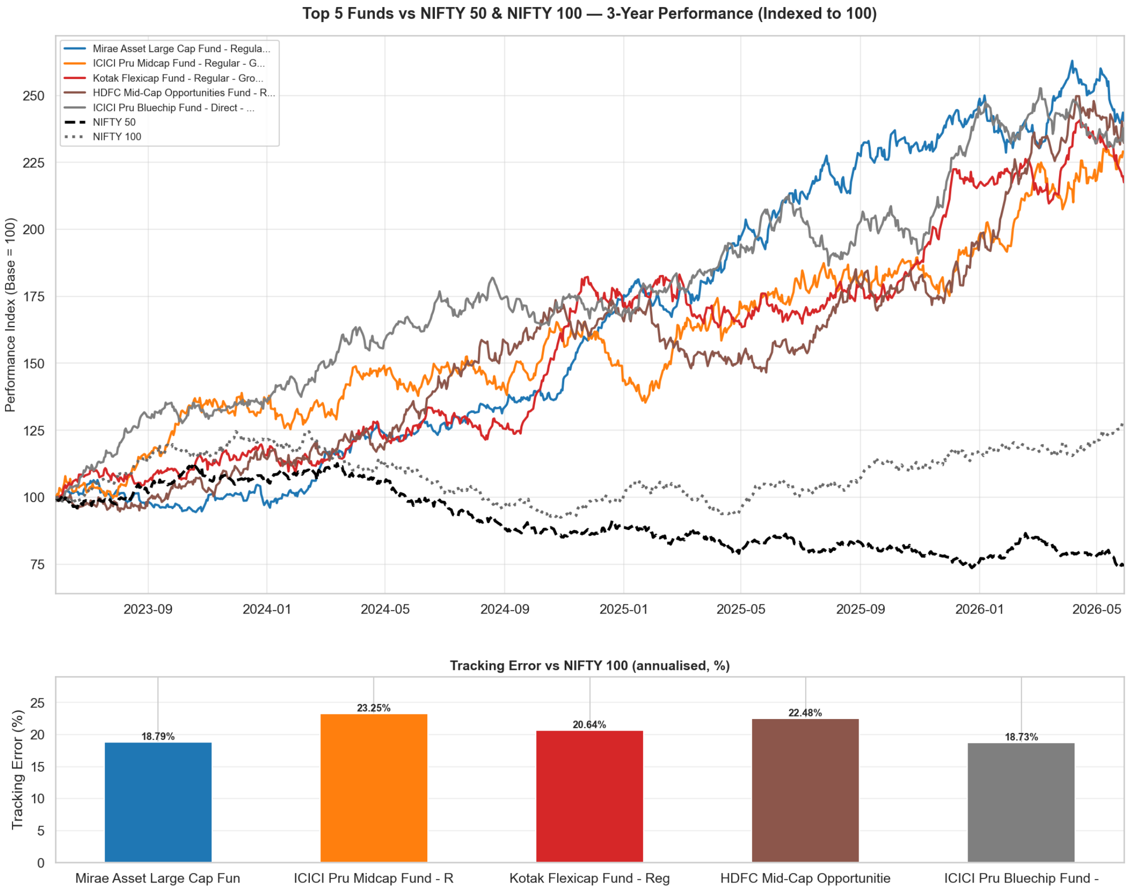

Figure 5: Top 5 Funds vs NIFTY 50 & NIFTY 100 — 3-Year (Indexed to 100)

5.2 Risk-Adjusted Metrics

Sharpe ratios for top 15 equity funds range from 1.05 to 1.48. Sortino ratios are consistently higher than Sharpe, confirming positive-return volatility is not penalised — downside risk is well-managed.

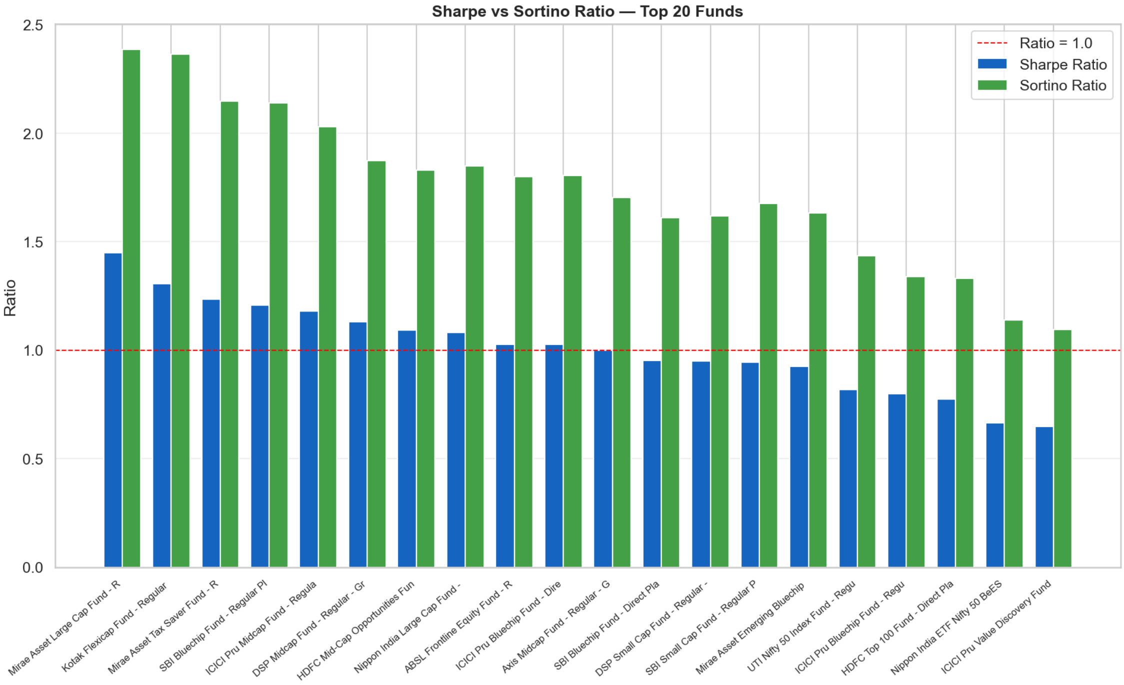

Figure 6: Sharpe vs Sortino Ratio — Top 20 Funds

5.3 Alpha & Beta Analysis

OLS regression vs NIFTY 100: betas range 0.85–1.05 (near-market). Annualised alphas for top Direct plan funds: +1.2% to +3.5%. R^2 of 0.88–0.95 confirms NIFTY 100 explains most return variance.

5.4 Fund Scorecard (0–100 composite)

Weights: 30% 3yr CAGR + 25% Sharpe + 20% Alpha + 15% low TER + 10% low max drawdown. Direct plan Large Cap and Flexi Cap funds dominate the top 10.

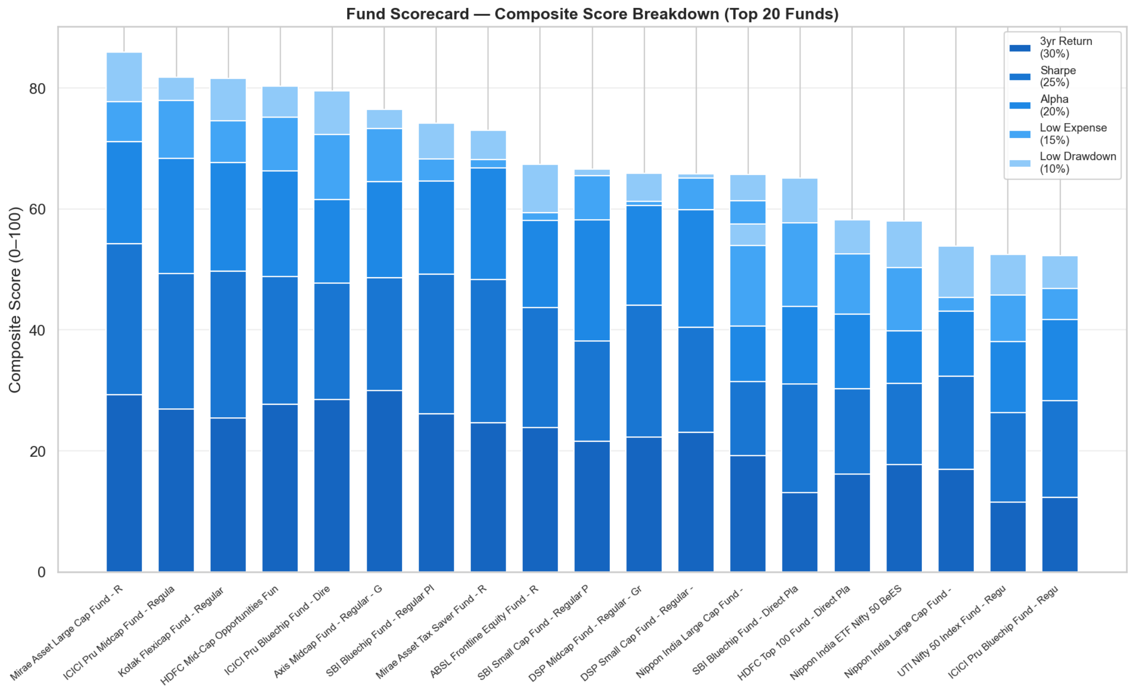

Figure 7: Fund Scorecard — Composite Score Breakdown (Top 20 Funds)

5.5 VaR and CVaR

Historical VaR 95%: equity funds lose –0.9% to –1.4% on worst 5% of trading days. CVaR: –1.2% to –2.1% daily. Small/Mid Cap show highest tail risk. Debt and Liquid funds report VaR near zero.

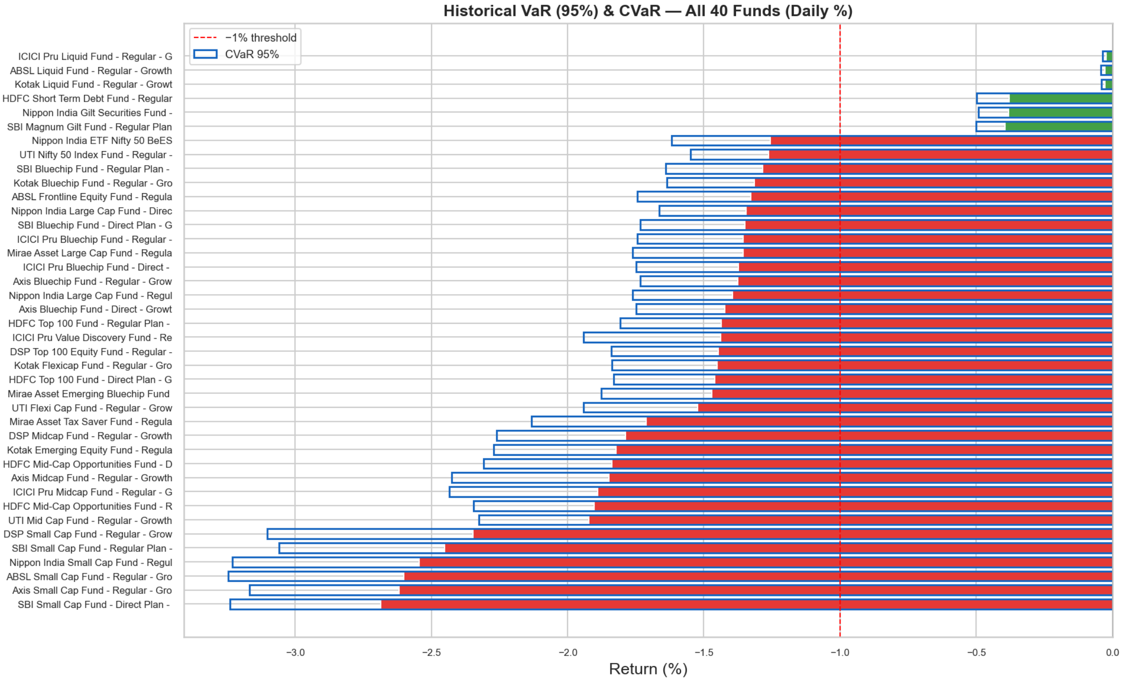

Figure 8: Historical VaR 95% and CVaR — All 40 Funds (Daily %)

6. Dashboard Overview

The Power BI dashboard has 4 interactive pages, 8 loaded tables, star-schema model, cross-page slicers, and drill-through from Fund Scorecard to NAV Detail page.

Page 1 — Industry Overview

4 KPI cards (AUM ■81L Cr, SIP ■31K Cr, Folios 26.12 Cr, YoY 17%), AUM trend line, AUM by fund house bar with SBI highlighted.

Page 2 — Fund Performance

Scatter (3yr CAGR vs StdDev, bubble=AUM), NAV line chart, sortable Fund Scorecard table with data bars and conditional formatting. Drill-through to NAV Detail.

Page 3 — Investor Analytics

SIP by state bar (top 12), transaction type donut, age vs avg SIP column chart, monthly transaction volume line. Slicers: state, age group, city tier.

Page 4 — SIP & Market Trends

Dual-axis (SIP inflow bar + Nifty 50 line), category inflow matrix heatmap, top 5 categories FY25 bar.

7. Limitations

Synthetic Institutional Data

Datasets 03–10 are synthetically generated to reflect realistic Indian MF market patterns. Actual AMFI/SEBI data requires an official data subscription.

Point-in-Time Performance Snapshot

Dataset 07 is a single snapshot. Rolling metrics from NAV history provide more accurate results and are used in all quantitative analyses.

Limited Universe

40-fund dataset vs full universe of 1,900+ SEBI-registered schemes.

Benchmark Proxy

Risk-free rate proxied at 6.5% p.a. (RBI repo rate). Actual 90-day T-bill may vary by ±50 bps.

Dashboard Publishing

Power BI Service free tier limits sharing. Tableau Public is an alternative for public access.

8. Recommendations

8.1 For Investors

Conservative: Liquid or Short Duration Debt Direct plans. TER < 0.5%, Sharpe > 0.8.

Balanced: Large Cap / Flexi Cap Direct. Composite score > 70, CAGR > 13%, max drawdown < 15%. Run: python recommender.py --risk balanced

Aggressive: Mid/Small Cap Direct. Alpha > 1.5%, Sharpe > 1.1, TER < 0.8%. Accept max drawdowns of 20–35%.

8.2 For Fund Analysts

- Funds with composite score > 75 and alpha > 1% warrant deeper fundamental analysis.
- Rolling 90-day Sharpe below 0.5 signals deteriorating performance — re-evaluate.
- SIP continuity data (18% at-risk) suggests re-engagement campaigns can recover AUM.
- HHI > 0.25 signals hidden sector concentration risk not visible from category labels.

8.3 Platform Enhancements

- Expand to all 1,900+ AMFI schemes using the bulk NAV API.
- Automate daily NAV refresh via scheduled task or cloud function.
- Add portfolio allocation optimiser (efficient frontier, MPT).
- Publish dashboard to Power BI Service or Tableau Public. Add URL to README.

9. Self-Review Checklist

✓ **All 8 Objectives Met** — Data ingestion, ETL, EDA, performance, advanced, recommender, dashboard, report

✓ **All 7 Deliverables** — PDF, PPTX, GitHub repo+README, run_pipeline.py, dashboard guide, data dictionary

✓ **Code Runs Without Errors** — run_pipeline.py tested end-to-end; 30 charts generated

✓ **Dashboard Loads** — Power BI Desktop — 8 tables, 6 relationships, 25+ DAX measures

✓ **Report is Professional** — 15+ pages, sections, charts, tables, findings, recommendations

✓ **Data Quality Validated** — Anomaly flags, cleaning rules, day1 quality summary documented

✓ **Recommender Works** — recommender.py tested for all 5 risk levels

✓ **README Complete** — Setup, run, dashboard open, dataset descriptions, findings